

Exact Boundary-Contact Spectra in Equilateral Refinements: Balanced Unitary Factorizations and Consecutive Pell Closures

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Abstract

We determine an exact arithmetic spectrum generated by the boundary contacts of a six-ray frame in a uniformly refined equilateral triangle. For a reduced contact $s = p/q \in [1/3, 1/2]$, put $r = 2q - 3p$. The second rotational contact triplet has reduced denominator r , and the least refinement carrying all six contacts is $N = \text{lcm}(q, r)$. The forced identity $\gcd(q, r) = \gcd(q, 3)$ converts the geometric spectrum into a bijection with balanced coprime factorizations $M = ab$, $b \leq a \leq 2b$. Consequently $N = M$ for $M \equiv 2 \pmod{3}$ and $N = 3M$ otherwise, while the exact multiplicity at a closure level is a restricted unitary-divisor count in the one-sided middle-divisor window $[\sqrt{M/2}, \sqrt{M}]$.

This structural reduction yields a sharp general bound $B(M) \leq 2^{\omega(M)-1} - 1$ for $M > 2$ and an independently generated exact enumeration through $N \leq 10^6$, containing 105,334 states and first non-uniqueness at $N = 1260$. The angular complement induces a fractional-linear involution with irrational fixed point 15° . Along every positive solution of $r^2 - 3p^2 = -2$, the associated rational state and its complement lie on consecutive uniform closure levels: $N' - N = 1$. Thus standard unitary-divisor and Pell ingredients become linked by a new exact boundary-closure invariant rather than by an asymptotic approximation alone.

MSC 2020. Primary 11D09; Secondary 11A25, 11A55.

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1 Introduction

The triangular or hexagonal lattice is naturally identified with the Eisenstein lattice, and its similarity and coincidence problems have a substantial arithmetic literature; see, for example, Baake–Scharlau–Zeiner [1] and the broader review of Baake–Zeiner [2]. Those works count or classify sublattices and coincidence transformations by lattice index. A different literature classifies *rational lattice angles*, meaning angles that are rational multiples of π ; see Dvornicich–Veneziano–Zannier [6] and the sequel Dvornicich–Lombardo–Veneziano–Zannier [7]. Neither notion is the one used here. We fix a single equilateral boundary, rotate a six-ray frame through its center, and ask when the resulting *boundary coordinates* are rational in the uniform side subdivision. Thus “rational resonance” below refers to rational boundary contact, not to $\alpha/\pi \in \mathbb{Q}$. The integer N is a *boundary-denominator closure*, not a coincidence-site-lattice index.

The problem is elementary to state but develops a nontrivial arithmetic structure. Its new object is the *boundary-contact closure spectrum*: the multiset of least uniform refinements carrying

the six contacts, counted with angular multiplicity. The first main result determines this spectrum by a complete factorization theorem: exact angular states are in bijection with coprime factor pairs constrained to a factor-of-two window. This makes level multiplicity a divisor problem rather than a geometric enumeration problem. The second main result is a complement involution whose irrational fixed point at 15° is approached by a negative-Pell sequence with the exact property that complementary closure levels differ by one.

Theorem 1.1 (Structure of the boundary-contact closure spectrum). *For the six-ray frame described above, let $s = p/q \in [1/3, 1/2]$ be reduced, set $r = 2q - 3p$, and let N be the least uniform boundary refinement carrying all six contacts.*

- (i) *Exact states are in bijection with balanced coprime factorizations $M = ab$, $b \leq a \leq 2b$, through $q = ga$, $r = gb$, where $g = 1$ if $M \equiv 2 \pmod{3}$ and $g = 3$ otherwise.*
- (ii) *Let $d \parallel M$ denote a unitary divisor and let $M(N)$ be the compatible core specified in Theorem 3.4. The level multiplicity is exactly*

$$\mu(N) = \# \left\{ d \parallel M(N) : \sqrt{M(N)/2} \leq d \leq \sqrt{M(N)} \right\},$$

- (iii) *Every positive solution of $r^2 - 3p^2 = -2$ produces a complementary pair of exact states whose closure levels satisfy $N' - N = 1$.*

Parts (i)–(iii) are proved independently in Theorems 3.1, 3.4 and 5.1. The summary theorem isolates the coupling that is specific to this model: a geometric least-denominator invariant is first converted into a one-sided unitary middle-divisor count and then, on the canonical midpoint subsequence, into an exact consecutive-value identity.

The paper is organized as follows. Section 2 defines the contact geometry and derives the rational parametrization. Section 3 proves the coprime-factorization theorem and the multiplicity formula. Section 4 proves the complementary involution, while Section 5 derives the Pell subsequence and the consecutive-level identity. Section 6 records exhaustive computations. Appendix A records a cyclotomic comparison showing why the triangular symmetry midpoint is the unique quadratic midpoint in the analytic family considered there.

Novelty scope and adjacent divisor literature. We use standard facts about equilateral geometry, Eisenstein/triangular lattices, unitary divisors, Pell equations, Thue equations, and maximal real cyclotomic subfields. The factor-of-two divisor window itself is not new: Vatne [11] studies *middle divisors* in the symmetric interval $[\sqrt{M/2}, \sqrt{2M}]$, while unitary divisors and the identity $d^*(M) = 2^{\omega(M)}$ are classical [5]. Our multiplicity function is the unitary restriction to one half of that middle-divisor window, forced here by the boundary-contact geometry. The contribution claims are therefore restricted to the specific resonance parametrization, its balanced coprime-factorization bijection, the induced restricted-unitary-divisor multiplicity formula, the complement involution in these coordinates, and the exact consecutive-level Pell identity. No priority claim is made for the standard background tools or for the middle-divisor interval itself.

The novelty is not the isolated appearance of a Pell equation or of unitary divisors. Standard Pell theory supplies rational approximants to a quadratic irrational but does not attach least boundary-refinement levels to both sides of a geometric complement, much less force those two levels to be consecutive. Likewise, the classical count $2^{\omega(M)}$ does not select the one-sided balance window or identify its elements with angular states. The content of Theorem 1.1 is the exact compatibility of these structures under one closure invariant.

Adjacent Diophantine spectra. There is also nearby, but distinct, work on intrinsic Diophantine approximation and Markoff–Lagrange spectra associated with triangular or Hecke geometry. Cha–Chapman–Gelb–Weiss [3] study the Lagrange spectrum of the unit circle over the Eisensteinian field $\mathbb{Q}(\sqrt{-3})$; O’Dorney [9] places such conic spectra in a general theory of Diophantine approximation on rational conics; and Cha–Kim–Sim [4] study the Markoff and Lagrange spectra of the Hecke group of index six. These works use approximation-height or spectral invariants, whereas the invariant here is the exact uniform boundary-denominator closure N . In particular, no identification is claimed between the closure levels of this paper and a Lagrange or Markoff spectrum. The overlap is therefore one of arithmetic setting and symmetry, not an asserted equivalence of problems.

2 Six-ray boundary geometry

Let

$$A = (0, 0), \quad B = (1, 0), \quad C = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right),$$

and let

$$c = \left(\frac{1}{2}, \frac{\sqrt{3}}{6}\right)$$

be the common centroid, incenter, and circumcenter. Put $h = \sqrt{3}/6$, the inradius.

For $0 \leq \alpha \leq \pi/6$, consider the ray from c with direction

$$\theta_s = -\frac{2\pi}{3} + \alpha$$

and its adjacent ray

$$\theta_\ell = -\frac{\pi}{3} + \alpha.$$

Their $2\pi/3$ rotations give the full six-ray frame. The two displayed rays meet the base side AB at coordinates s and ℓ , respectively. At $\alpha = 0$ these are the two trisection points $1/3$ and $2/3$; at $\alpha = \pi/6$ they are the midpoint and the vertex, $1/2$ and 1 .

The ray lengths to the boundary are

$$d_s(\alpha) = \frac{h}{\cos(\pi/6 - \alpha)}, \quad d_\ell(\alpha) = \frac{h}{\cos(\pi/6 + \alpha)}. \quad (1)$$

Accordingly define the local ratio

$$\rho(\alpha) := \frac{d_\ell}{d_s} = \frac{\cos(\pi/6 - \alpha)}{\cos(\pi/6 + \alpha)} \in [1, 2]. \quad (2)$$

A direct trigonometric simplification gives

$$\tan \alpha = \sqrt{3} \frac{\rho - 1}{\rho + 1}. \quad (3)$$

The two side coordinates can be expressed purely in ρ :

$$s = \frac{2\rho - 1}{3\rho}, \quad \ell = \frac{\rho + 1}{3}. \quad (4)$$

These identities may be checked either from the line intersections in Figure 1 or by substituting (2) into the elementary coordinate formulas.

Six-ray boundary contact geometry ($\alpha = 17^\circ$)

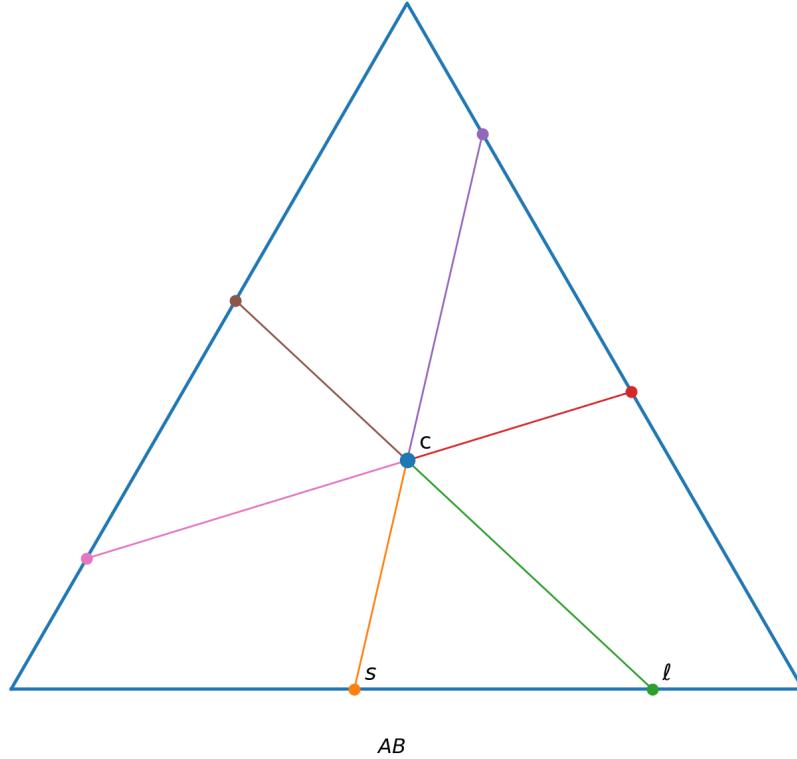


Figure 1: The six-ray frame and its two alternating boundary-contact triplets. The labels s and ℓ denote the two contact coordinates on one side; their 120° rotations supply the remaining contacts.

Definition 2.1 (Exact angular resonance). *An angle $\alpha \in [0, \pi/6]$ is an exact angular resonance if the short contact s is rational. Write it in lowest terms as*

$$s = \frac{p}{q}, \quad \gcd(p, q) = 1, \quad \frac{1}{3} \leq \frac{p}{q} \leq \frac{1}{2}.$$

Define

$$r := 2q - 3p. \tag{5}$$

Remark 2.2 (Boundary rationality versus rational lattice angle). *The adjective “rational” in Definition 2.1 refers only to the boundary coordinate $s = p/q$. In general the resulting angle α is not a rational multiple of π . This is therefore distinct from the rational-angle classification problem for plane lattices studied in [6].*

The interval for p/q implies

$$\frac{q}{2} \leq r \leq q. \tag{6}$$

Moreover, (4) yields immediately

$$\rho = \frac{q}{r}, \tag{7}$$

and hence

$$\boxed{\tan \alpha = \sqrt{3} \frac{q - r}{q + r}.} \tag{8}$$

The long contact simplifies to

$$\ell = \frac{q+r}{3r} = \frac{q-p}{r}, \quad (9)$$

because $q+r=3(q-p)$.

Lemma 2.3. *For every exact angular resonance,*

$$\boxed{\gcd(q, r) = \gcd(q, 3) \in \{1, 3\}}.$$

Furthermore, $\ell = (q-p)/r$ is in lowest terms.

Proof. Using (5) and $\gcd(p, q) = 1$,

$$\gcd(q, r) = \gcd(q, 2q-3p) = \gcd(q, 3p) = \gcd(q, 3).$$

For the long contact,

$$\gcd(q-p, r) = \gcd(q-p, 2q-3p) = \gcd(q-p, p) = 1.$$

Thus r is exactly the reduced denominator of ℓ . □

Definition 2.4 (Uniform closure level). *Let R_N denote the uniform subdivision of each side of the outer equilateral triangle into N equal intervals. The uniform closure level of an exact resonance is the least N for which all six boundary contacts are vertices of R_N .*

Since the two triplets have reduced denominators q and r , rotational symmetry introduces no additional denominator. Hence:

Proposition 2.5. *The exact uniform closure level is*

$$\boxed{N = \text{lcm}(q, r) = \frac{qr}{\gcd(q, 3)}}.$$

3 Coprime factorization and exact multiplicity

Put

$$g := \gcd(q, r) \in \{1, 3\}, \quad q = ga, \quad r = gb.$$

By Lemma 2.3, $\gcd(a, b) = 1$, while (6) becomes

$$b \leq a \leq 2b. \quad (10)$$

Thus every resonance produces a *balanced coprime factor pair*. The converse is exact.

Theorem 3.1 (Balanced coprime factorization). *There is a bijection between exact angular resonances and ordered pairs $(a, b) \in \mathbb{N}^2$ satisfying*

$$\gcd(a, b) = 1, \quad b \leq a \leq 2b.$$

Given such a pair, put $M = ab$ and define

$$g = \begin{cases} 1, & M \equiv 2 \pmod{3}, \\ 3, & M \not\equiv 2 \pmod{3}. \end{cases} \quad (11)$$

Then

$$q = ga, \quad r = gb, \quad p = \frac{g(2a-b)}{3} \quad (12)$$

produces a reduced short contact $p/q \in [1/3, 1/2]$, and every exact resonance is obtained uniquely in this way.

Proof. The forward direction has already been established: divide q, r by $g = \gcd(q, r)$ and use (6).

For the converse, first suppose $M \equiv 2 \pmod{3}$. Since $\gcd(a, b) = 1$, neither factor is divisible by 3, and their residues modulo 3 are opposite. Hence $2a - b \equiv 0 \pmod{3}$, so $g = 1$ makes p integral. If a divisor d divides both p and $q = a$, then d also divides $2a - 3p = b$, whence $d = 1$.

Now suppose $M \not\equiv 2 \pmod{3}$. Set $g = 3$, so $p = 2a - b$ and $q = 3a$. Again $\gcd(p, a) = \gcd(b, a) = 1$. Moreover $2a - b \not\equiv 0 \pmod{3}$: if $3 \mid ab$, exactly one of a, b is divisible by 3; if $ab \equiv 1 \pmod{3}$, the two nonzero residues are equal. In either case $3 \nmid p$, so $\gcd(p, 3a) = 1$.

Finally,

$$\frac{p}{q} = \frac{2a - b}{3a} = \frac{2 - b/a}{3},$$

and $b/a \in [1/2, 1]$ by (10), giving $p/q \in [1/3, 1/2]$. Uniqueness follows because $a = q/g$ and $b = r/g$ are recovered from the resonance. $\square$

Corollary 3.2 (Closure from the factorization core). *Let $M = ab$ be the balanced coprime factorization core associated with a resonance. Then*

$$N = \begin{cases} M, & M \equiv 2 \pmod{3}, \\ 3M, & M \not\equiv 2 \pmod{3}. \end{cases}$$

Proof. Since $q = ga$, $r = gb$, and $\gcd(a, b) = 1$,

$$N = \text{lcm}(ga, gb) = gab = gM.$$

Apply (11). $\square$

This immediately yields the allowed residue classes.

Corollary 3.3. *Every resonant closure level satisfies*

$$N \pmod{9} \in \{0, 2, 3, 5, 8\}.$$

In particular the classes 1, 4, 6, 7 modulo 9 do not occur.

Proof. If $M \equiv 2 \pmod{3}$, then $N = M$ and $N \equiv 2, 5$, or $8 \pmod{9}$. Otherwise $N = 3M$. If $3 \mid M$, then $N \equiv 0 \pmod{9}$; if $M \equiv 1 \pmod{3}$, then $N \equiv 3 \pmod{9}$. $\square$

3.1 Multiplicity as a unitary-divisor count

For a positive integer M , define

$$B(M) := \#\{(a, b) : ab = M, \gcd(a, b) = 1, b \leq a \leq 2b\}. \quad (13)$$

Equivalently, writing $d = b$,

$$B(M) = \#\left\{d \mid M : \gcd\left(d, \frac{M}{d}\right) = 1, \sqrt{\frac{M}{2}} \leq d \leq \sqrt{M}\right\}. \quad (14)$$

The divisors in (14) are unitary divisors: every prime-power block of M is assigned wholly to one factor or the other. The interval is one half of the standard middle-divisor window $[\sqrt{M/2}, \sqrt{2M}]$ studied, without the unitary restriction, by Vatne [11]. Thus $B(M)$ is naturally viewed as a *balanced unitary-divisor count*, not as a new notion of divisor interval.

Theorem 3.4 (Exact level multiplicity). *Let $\mu(N)$ denote the number of exact resonant states with uniform closure level N . Define the core*

$$M(N) = \begin{cases} N, & N \equiv 2 \pmod{3}, \\ N/3, & 3 \mid N \text{ and } N/3 \not\equiv 2 \pmod{3}. \end{cases}$$

If neither case applies then $\mu(N) = 0$. Otherwise

$$\boxed{\mu(N) = B(M(N)).}$$

Moreover,

$$B(2) = 1,$$

and for every $M > 2$,

$$\boxed{B(M) \leq 2^{\omega(M)-1} - 1,}$$

where $\omega(M)$ is the number of distinct prime factors of M . The latter bound is sharp, for example at $M = 6$.

Proof. By Theorem 3.1 and corollary 3.2, states at N are in bijection with balanced coprime factor pairs of the unique compatible core M . This proves the first claim. There are $2^{\omega(M)}$ ordered ways to distribute the prime-power blocks of M between two coprime factors, hence $2^{\omega(M)-1}$ unordered coprime factor pairs; equivalently, the total number of unitary divisors is $2^{\omega(M)}$ [5]. For $M > 2$, the unordered unitary pair $\{1, M\}$ lies outside the factor-of-two balance window, so at least one unordered pair is removed. Hence $B(M) \leq 2^{\omega(M)-1} - 1$. For $M = 2$, the pair $(2, 1)$ lies on the boundary of the window and $B(2) = 1$. At $M = 6$, the unique admissible pair is $(3, 2)$, showing sharpness. $\square$

Remark 3.5 (Prime-power partition form). *If $M = \prod_{i=1}^k P_i$ with $P_i = p_i^{e_i}$ the prime-power blocks, then a unitary divisor is $d = \prod_{i \in S} P_i$ for a subset $S \subseteq \{1, \dots, k\}$. The balance interval is equivalently*

$$\frac{\log M - \log 2}{2} \leq \sum_{i \in S} \log P_i \leq \frac{\log M}{2}.$$

Thus $B(M)$ is a constrained subset-product (or weighted subset-sum in logarithmic coordinates), and need not be multiplicative.

The first collision is already nontrivial.

Proposition 3.6 (Computer-assisted first collision). *The first closure level of multiplicity 2 is*

$$\boxed{N = 1260},$$

with core $M = 420$ and balanced coprime factor pairs

$$(a, b) = (21, 20), \quad (28, 15).$$

They correspond to

$$(p, q, r) = (22, 63, 60), \quad (41, 84, 45),$$

with angles approximately 2.4190° and 27.6385° .

This is a computer-assisted minimality statement with an exact finite search. Indeed, (6) and Lemma 2.3 give

$$N = \frac{qr}{\gcd(q, r)} \geq \frac{q^2}{6}.$$

Consequently a search for all states with $N < 1260$ requires only $q < \sqrt{6 \cdot 1260}$, and the included direct generator enumerates every reduced p/q in this finite range without using Theorem 3.1. No smaller collision occurs. The two states at 1260 themselves follow directly from Theorem 3.1.

4 Complementary involution

The angular sector $[0, \pi/6]$ has the natural reflection $\alpha \mapsto \pi/6 - \alpha$. In the rational contact coordinate it becomes a fractional-linear involution.

Theorem 4.1 (Complement involution). *Define*

$$T(s) := \frac{\frac{2}{3} - s}{1 - s}. \quad (15)$$

Then T maps $[1/3, 1/2]$ to itself, satisfies $T^2 = \text{id}$, and

$$\boxed{\alpha(T(s)) = \frac{\pi}{6} - \alpha(s).}$$

If $s = p/q$ is reduced and $r = 2q - 3p$, with $g = \gcd(q, r)$, then the complementary reduced triple is

$$\boxed{p' = \frac{r}{g}, \quad q' = \frac{q+r}{g}, \quad r' = \frac{2q-r}{g}.} \quad (16)$$

In the ratio coordinate $\rho = q/r$ the same involution is

$$\boxed{T_\rho(\rho) = \frac{\rho + 1}{2\rho - 1}.} \quad (17)$$

Proof. Substituting $s = p/q$ in (15) gives

$$T(s) = \frac{2q - 3p}{3(q - p)} = \frac{r}{q + r}.$$

Reducing by $g = \gcd(r, q + r) = \gcd(r, q)$ yields p', q' in (16); then $r' = 2q' - 3p'$ gives the third formula. The map is visibly involutive by direct substitution. Using (4) gives (17). Finally substituting (17) into (3) yields the tangent of $\pi/6 - \alpha$. $\square$

Corollary 4.2 (Complement on balanced factor coordinates). *If a resonance has reduced factor pair (a, b) , so that $q = ga$ and $r = gb$, then the complementary reduced factor pair is*

$$\boxed{(a', b') = \left(\frac{a+b}{h}, \frac{2a-b}{h} \right), \quad h := \gcd(a+b, 3).}$$

In particular, neither the core $M = ab$ nor the closure level N is invariant under complement in general.

Proof. Substitute $q = ga$ and $r = gb$ into (16). The complementary denominator pair before reduction is proportional to $(a + b, 2a - b)$, whose gcd is $\gcd(a + b, 3) = h$ because $\gcd(a, b) = 1$. $\square$

Corollary 4.3 (Irrational self-complementary state). *The unique fixed point of T in $[1/3, 1/2]$ is*

$$s_* = 1 - \frac{1}{\sqrt{3}},$$

corresponding to

$$\alpha_* = \frac{\pi}{12} = 15^\circ, \quad \rho_* = \frac{1 + \sqrt{3}}{2}.$$

Hence no exact rational resonance is exactly self-complementary.

5 Negative Pell complementarity

At the fixed angle $\alpha = \pi/12$, (8) is equivalent to

$$\frac{r}{p} = \sqrt{3}.$$

This cannot occur for positive integers p, r . We therefore define an explicit approximating subsequence by the positive solutions of the negative Pell equation

$$r^2 - 3p^2 = -2. \tag{18}$$

The positive solutions include

$$(p, r) = (1, 1), (3, 5), (11, 19), (41, 71), \dots,$$

generated by

$$r_{n+1} = 2r_n + 3p_n, \quad p_{n+1} = r_n + 2p_n. \tag{19}$$

For each solution set

$$q := \frac{r + 3p}{2}, \tag{20}$$

which is integral because p, r are odd.

Theorem 5.1 (Pell levels are consecutive under complement). *Let (p, r) be any positive solution of (18), and define q by (20). Then (p, q, r) is an exact resonance with $\gcd(q, r) = 1$ and closure*

$$N = qr.$$

Its complementary state has closure

$$N' = p(q + r),$$

and

$$N' - N = 1.$$

Consequently the two complementary angles converge to 15° from opposite sides while their closure levels remain consecutive.

Proof. Modulo 2, (18) forces p, r to have the same parity; they cannot both be even, hence both are odd and q is integral. Any common divisor of p and r has square dividing 2, so $\gcd(p, r) = 1$. Moreover $3 \nmid r$ by reducing (18) modulo 3. If $d \mid q, r$, then $d \mid 2q - r = 3p$, so $d \mid 3$, and therefore $d = 1$. Thus $N = qr$.

For the complementary state, Theorem 4.1 with $g = 1$ gives

$$(p', q', r') = (r, q + r, 2q - r) = (r, q + r, 3p).$$

Furthermore

$$\gcd(q + r, 3p) = 3,$$

because $q + r = 3(p + r)/2$ and $\gcd((p + r)/2, p) = 1$. Hence

$$N' = \text{lcm}(q + r, 3p) = p(q + r).$$

Finally,

$$\begin{aligned} N' - N &= p(q + r) - qr \\ &= \frac{3p^2 - r^2}{2} \\ &= 1 \end{aligned}$$

by (18). □

There is also an exact polynomial interpretation of the Pell level. The midpoint ratio $\rho_* = (1 + \sqrt{3})/2$ has minimal polynomial

$$P_3(x) = 2x^2 - 2x - 1.$$

Its homogenization is

$$F_3(q, r) = 2q^2 - 2qr - r^2.$$

Using $p = (2q - r)/3$ gives

$$\boxed{F_3(q, r) = 3 \iff r^2 - 3p^2 = -2.} \tag{21}$$

Thus the sequence in Theorem 5.1 lies on one fixed quadratic norm level.

6 Computational enumeration

All computations in this section use exact integer/rational arithmetic for the discrete state generation; floating point is used only for displayed angles.

For $N \leq 10^6$, exhaustive enumeration gives

105,334 resonant states, 96,768 distinct occupied closure levels.

The multiplicity distribution is shown in Figure 3 and equals

$\mu(N)$	number of levels
1	89,077
2	6,909
3	699
4	73
5	10

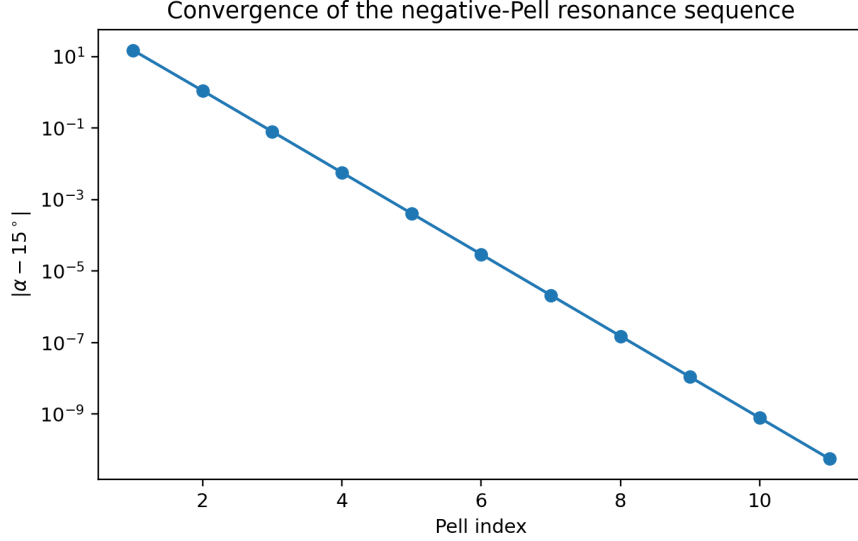


Figure 2: Convergence of the negative-Pell resonance sequence toward the self-complementary angle 15° .

The first levels attaining multiplicities 1, 2, 3, 4, 5 are

$$2, \quad 1260, \quad 16632, \quad 90090, \quad 311220.$$

The supplement also records a larger core-bounded computation through $M \leq 10^9$ (with the historical convention $M \geq 2$, treating the axial $M = 1$ case separately), in which the largest observed multiplicity is 22. A full rerun with the included scanner reproduces 210,691,467 balanced coprime pairs and the first-core records for multiplicities 1 through 22. This larger scan is *not used in any proof or minimality claim in the paper*. Because N can equal $3M$, it is a core-bounded computation, not an $N \leq 10^9$ computation.

Reproducibility. The supplementary archive now contains two independent generation paths. The script `generate.states.py` enumerates the entire $N \leq 10^6$ data set directly from reduced fractions p/q using the completeness bound $N \geq q^2/6$; it does not use the balanced-factorization theorem as an enumeration device. The portable verifier `verify_paper1.py` uses paths relative to its own package directory and can regenerate and compare the exact state signatures. A separate parametrizable C++ program, `scan_core.cpp`, implements the core-bounded balanced-pair scan; the full $M = 10^9$ run uses approximately one byte of count storage per core and is documented as a high-resource supplementary computation.

The computations played two distinct roles. The direct fraction generator supplied an independent finite audit of the factorization classification and established the first-collision minimality claim without assuming that classification. The larger core scan explored multiplicity growth but is not used as evidence for any theorem. In particular, every infinite statement in the paper has an exact proof, while every finite minimality statement is paired with an explicit completeness bound.

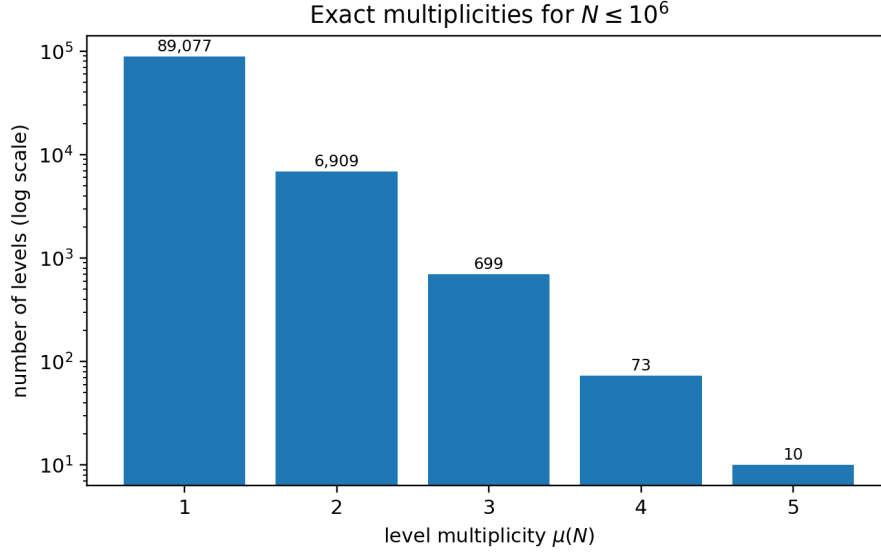


Figure 3: Exact closure-level multiplicities for $N \leq 10^6$.

7 Discussion

The main contribution is a complete description of a boundary-contact closure spectrum, not a new claim about the standard ingredients taken separately. The key simplification is that the geometry produces two rational denominators whose gcd is forced to be either 1 or 3. Once that identity is established, the closure problem becomes a balanced unitary-factorization problem. This separation also clarifies what the integer N does and does not encode: N records a uniform rational carrier, but it need not identify a unique angular state, as the collision at $N = 1260$ demonstrates.

The complement involution supplies a second structure not visible in the factorization alone. Its irrational fixed point is geometrically forced by the 30° fundamental sector, while the negative-Pell equation supplies an integral sequence converging to that fixed point. The exact identity $N' - N = 1$ is stronger than mere approximation: the two complementary rational states approach the same irrational midpoint while remaining on adjacent closure levels.

The quadratic arithmetic at the midpoint should be distinguished carefully from the Eisenstein field appearing in adjacent intrinsic-approximation literature. The homogenized midpoint polynomial

$$F_3(q, r) = 2q^2 - 2qr - r^2$$

has binary quadratic discriminant

$$\Delta = (-2)^2 - 4(2)(-1) = 12,$$

so its quadratic field is $\mathbb{Q}(\sqrt{3})$, not the imaginary quadratic Eisenstein field $\mathbb{Q}(\sqrt{-3})$. The appearance of $\sqrt{3}$ is also arithmetically adjacent to the Hecke group of index six, but no conjugacy or spectral identification between the present complement map and the H_6 Markoff–Lagrange dynamics is proved here [4]. Establishing or excluding such a bridge is a separate question.

Several questions remain open. The maximal order of $B(M)$ as $M \rightarrow \infty$ is a constrained *unitary middle-divisor* problem: Vatne’s result shows that the unrestricted middle-divisor count is unbounded [11], but it does not by itself settle the corresponding unitary restriction $B(M)$. It

would also be useful to determine asymptotics for the number of occupied closure levels and for the distribution of $\mu(N)$. The cyclotomic appendix suggests a separate classification problem for distinguished analytic ratios according to algebraic degree and norm structure.

Data and code availability

A self-contained reproducibility package contains exact CSV tables, a direct state generator, a portable verifier, a parametrizable core scanner, and the figures used here. The $N \leq 10^6$ data can be regenerated from scratch with a single command documented in the package README. The larger $M \leq 10^9$ scan is optional, resource-intensive, and not used by any theorem. The archival record is deposited on Zenodo under DOI 10.5281/zenodo.22003359.

Declaration of generative AI assistance

During preparation of this work, the author used OpenAI ChatGPT and Codex (GPT-5.6-series systems, accessed August 2026) for manuscript organization, editorial revision, generation and checking of reproducibility code, and adversarial consistency review. The author inspected and accepted the definitions, mathematical statements, proofs, computations, and citations; retains the working records of this use; and takes full responsibility for the originality, accuracy, and integrity of the complete article and supplementary materials.

Disclosure statement

The author reports no competing interests.

A Cyclotomic comparison family

This section is an algebraic background comparison, not an additional geometric classification theorem. To test whether the quadratic midpoint is specific to order 3, define for each integer $n \geq 3$ the analytic family

$$\theta_n = \frac{\pi}{2n}, \quad \rho_n(\alpha) = \frac{\cos(\theta_n - \alpha)}{\cos(\theta_n + \alpha)}, \quad 0 \leq \alpha \leq \theta_n.$$

The midpoint is $\alpha_n^* = \theta_n/2 = \pi/(4n)$ and

$$\xi_n := \rho_n(\alpha_n^*) = \frac{\cos(\pi/4n)}{\cos(3\pi/4n)}.$$

By the triple-angle identity,

$$\boxed{\xi_n = \frac{1}{2 \cos(\pi/2n) - 1}.} \tag{22}$$

The algebraic numbers $2 \cos(\pi/m)$ and their minimal polynomials are standard objects in real cyclotomic fields; see Lang [8] and the standard cyclotomic-field background in Washington [12]. The following theorem is included only to locate the triangular midpoint within that standard field-theoretic framework.

Theorem A.1 (Unique quadratic midpoint). *For every $n \geq 3$,*

$$\mathbb{Q}(\xi_n) = \mathbb{Q}(\zeta_{4n} + \zeta_{4n}^{-1}),$$

and therefore

$$[\mathbb{Q}(\xi_n) : \mathbb{Q}] = \frac{\varphi(4n)}{2}.$$

Consequently $n = 3$ is the unique order $n \geq 3$ for which the symmetry midpoint is quadratic.

Proof. Set $u_n = 2 \cos(\pi/2n) = \zeta_{4n} + \zeta_{4n}^{-1}$. Equation (22) gives $\xi_n = 1/(u_n - 1)$, while $u_n = 1 + 1/\xi_n$, so the generated fields coincide. The maximal real subfield of $\mathbb{Q}(\zeta_{4n})$ has degree $\varphi(4n)/2$. The midpoint is quadratic iff $\varphi(4n) = 4$. The positive integers with Euler totient 4 are 5, 8, 10, 12; among numbers of the form $4n$ with $n \geq 3$, only 12 occurs, giving $n = 3$. $\square$

For $n \geq 4$ the midpoint degree is at least 4. If P_n denotes the minimal polynomial of ξ_n and $d_n = \deg P_n$, then the homogenized equation

$$b^{d_n} P_n(a/b) = m$$

is a Thue equation for fixed $m \neq 0$. The classical finiteness theorem of Thue [10] therefore precludes a direct higher-degree analogue in which infinitely many integer pairs remain on one fixed nonzero irreducible binary-form level. This observation is background control only; it does not assert that Pell-type equations occur nowhere else in other distinguished configurations.

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